

Practical 4: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Key Questions:

1. How are DOM elements selected and modified?

Answer:

DOM elements are selected using JavaScript methods such as `getElementById()`, `querySelector()`, and `querySelectorAll()`. Once selected, their content, attributes, styles, or CSS classes are modified dynamically using properties like `innerHTML`, `textContent`, `style`, and `classList`. This allows the webpage to respond to user actions without reloading the page.

2. Are event listeners attached correctly?

Answer:

Yes. Event listeners are attached using the `addEventListener()` method, which is the recommended JavaScript approach.

- Different events such as click, submit, and change are handled efficiently.
- Event listeners are attached after the DOM has loaded to prevent errors.
- This keeps the JavaScript code separate from the HTML, making it cleaner and easier to maintain.
- Multiple events can be managed without affecting existing functionality.

3. Is `localStorage` used for remembering theme choice?

Answer:

Yes. The browser's `localStorage` is used to save the user's selected theme.

- When the user changes the theme, the selected value is stored in `localStorage`.
- Whenever the website is reopened or refreshed, the saved theme is retrieved automatically.
- This provides a personalized experience by maintaining the user's preference across sessions.
- Since the data is stored locally in the browser, no server interaction is required.

4. Does interactivity improve usability without breaking accessibility?

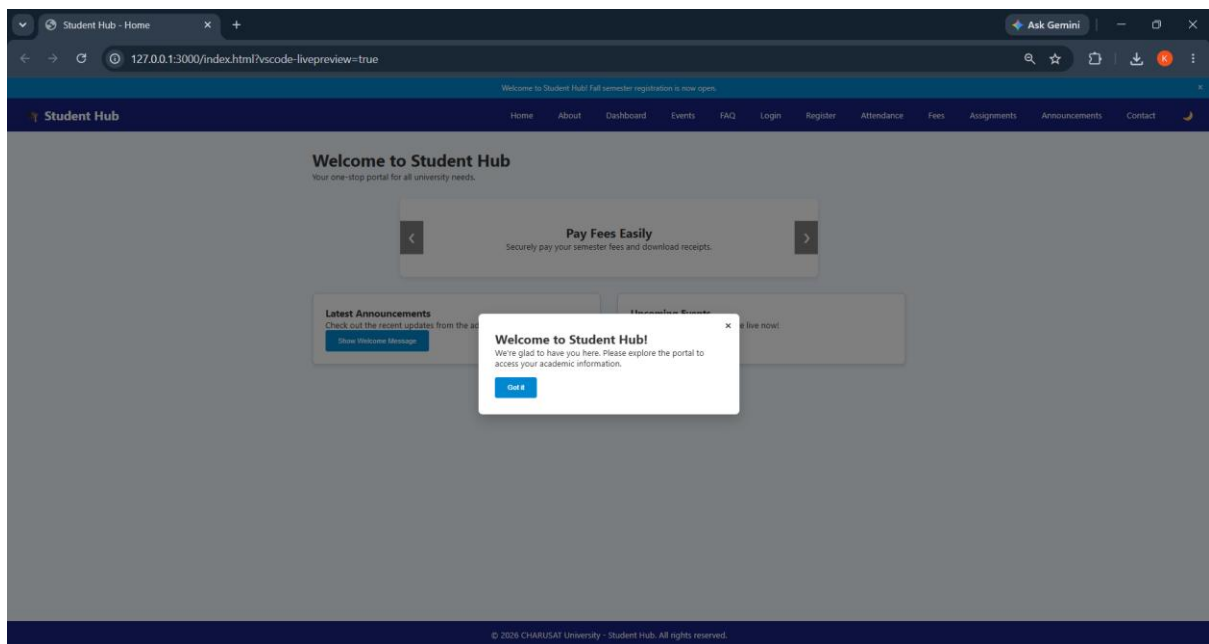
Answer:

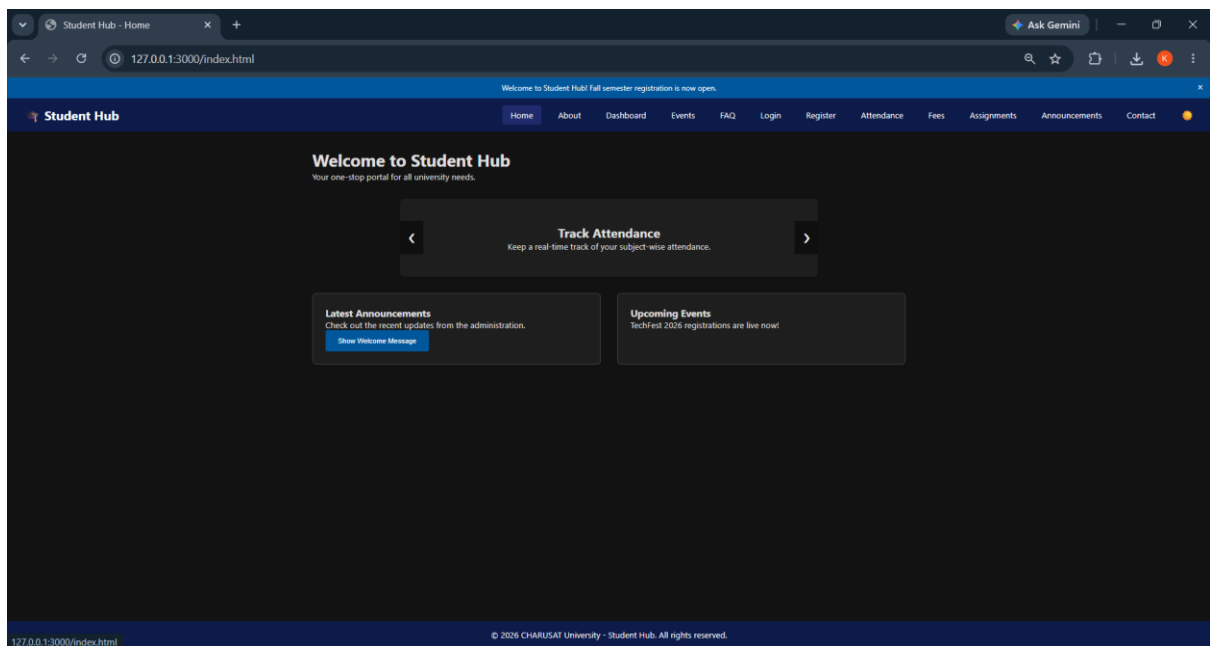
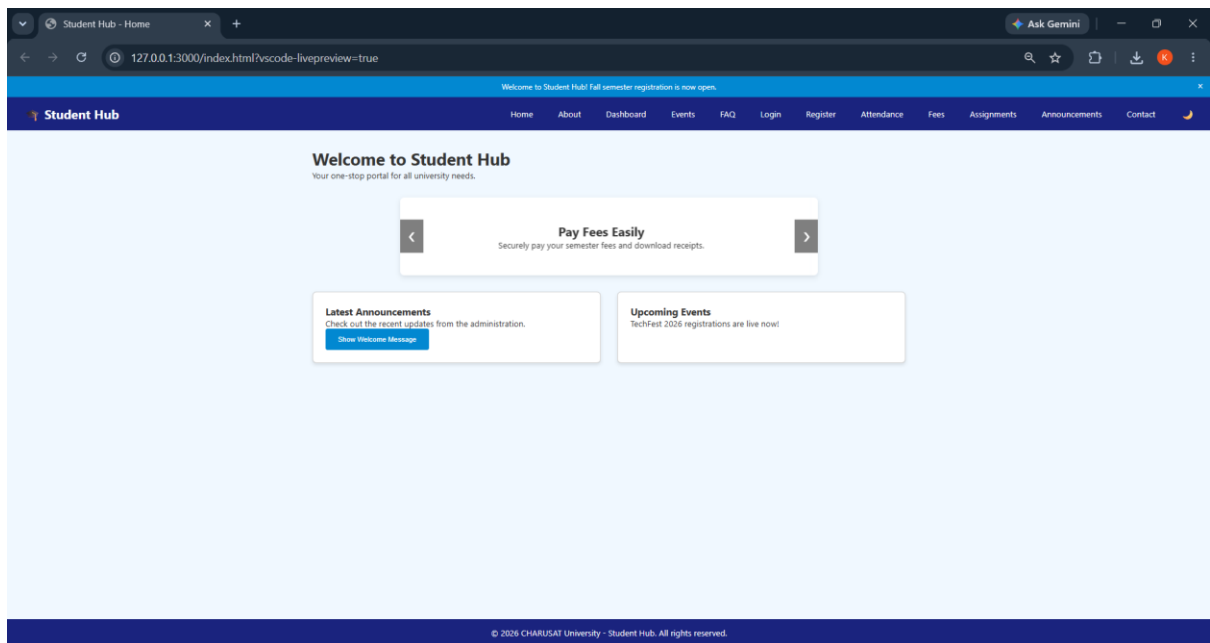
Yes. The interactive features are designed to improve the user experience while maintaining accessibility.

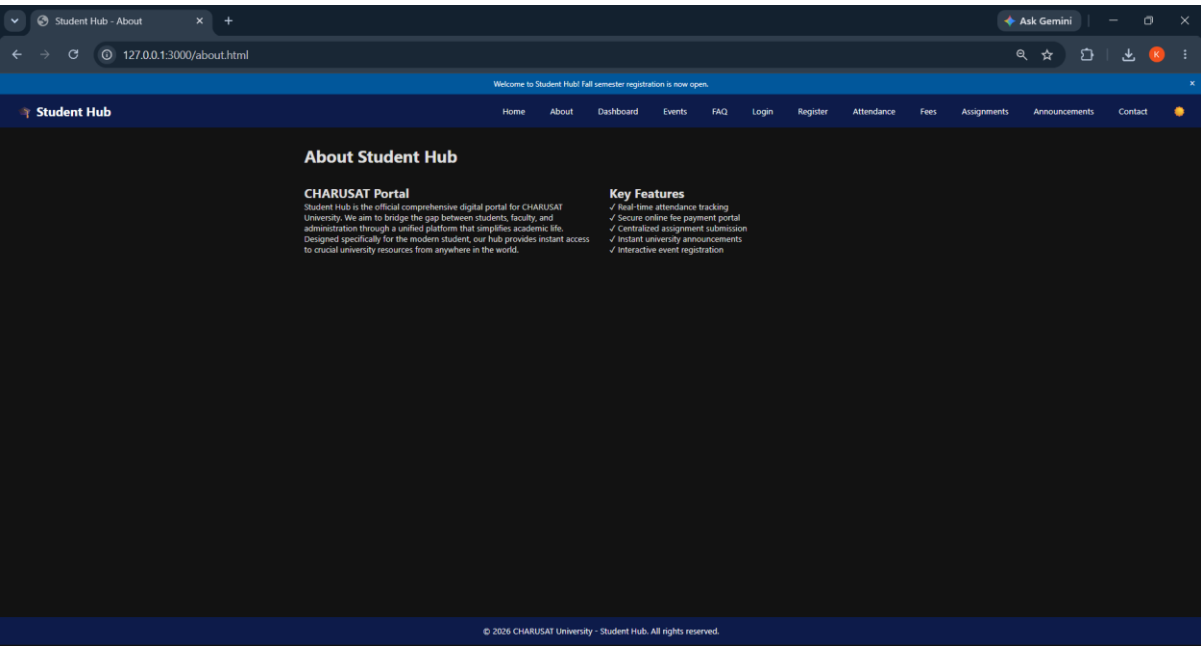
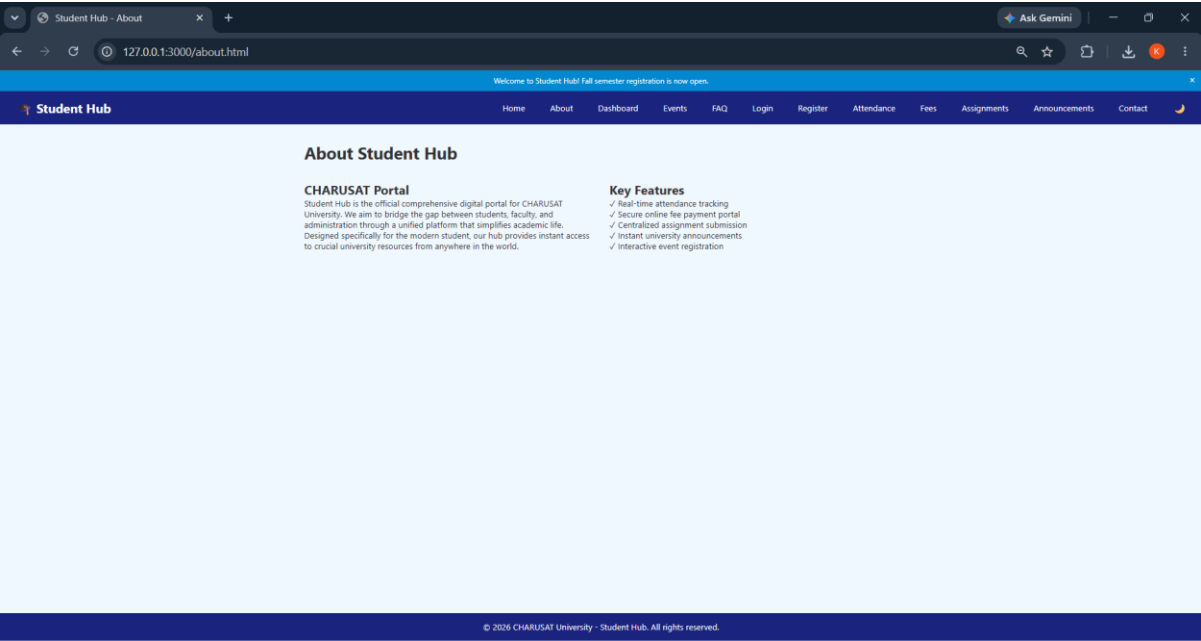
- Buttons and interactive elements are easy to identify and use.
- Proper HTML elements are used for better keyboard navigation.
- Good color contrast and readable text improve accessibility.
- Interactive features provide quick feedback to user actions without making the interface confusing.
- The design remains responsive and accessible across different devices and screen sizes.

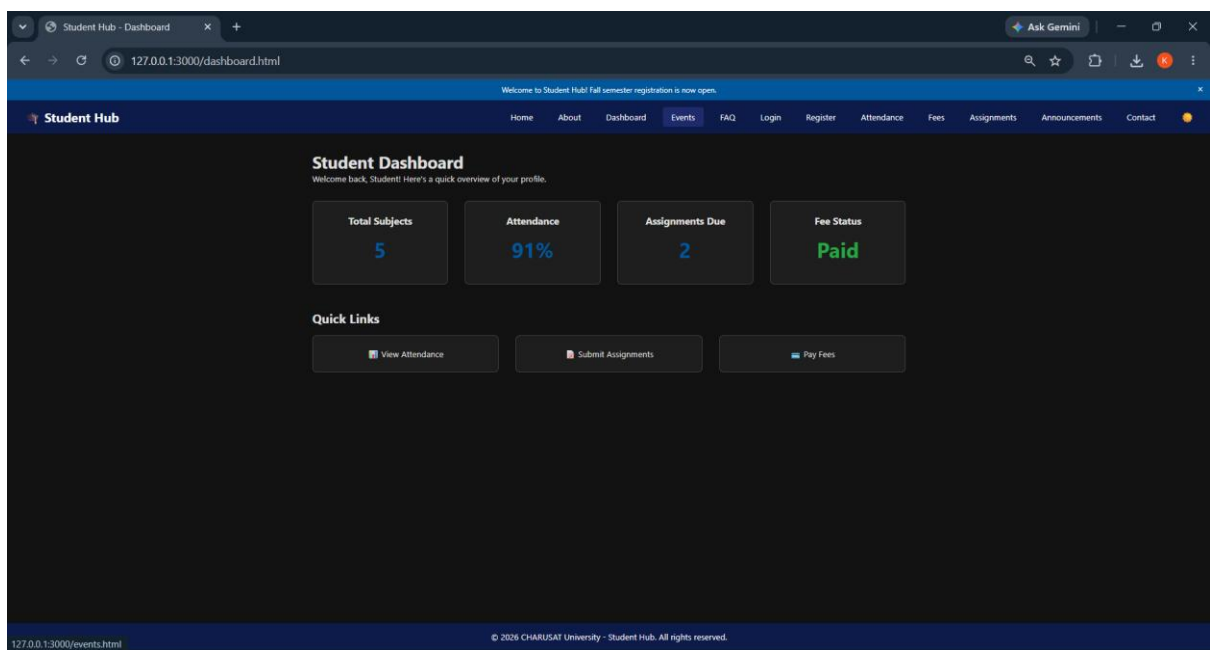
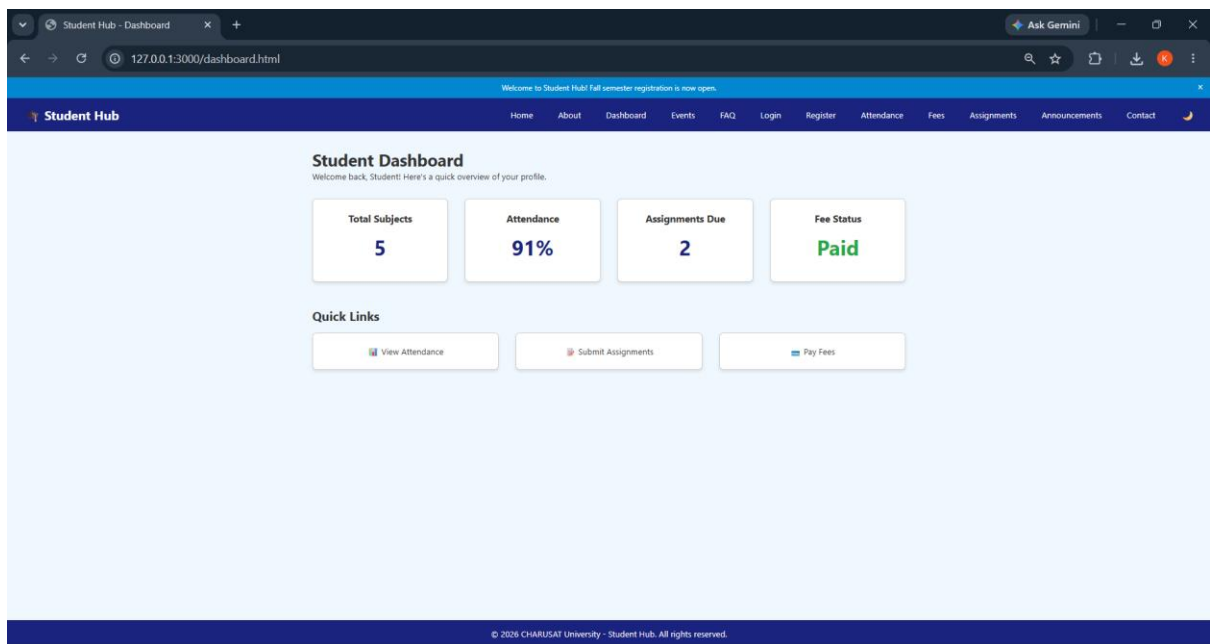
For Codes: <https://github.com/kunjborad21/WebDev-CSE1-A/tree/main/Practical%204>

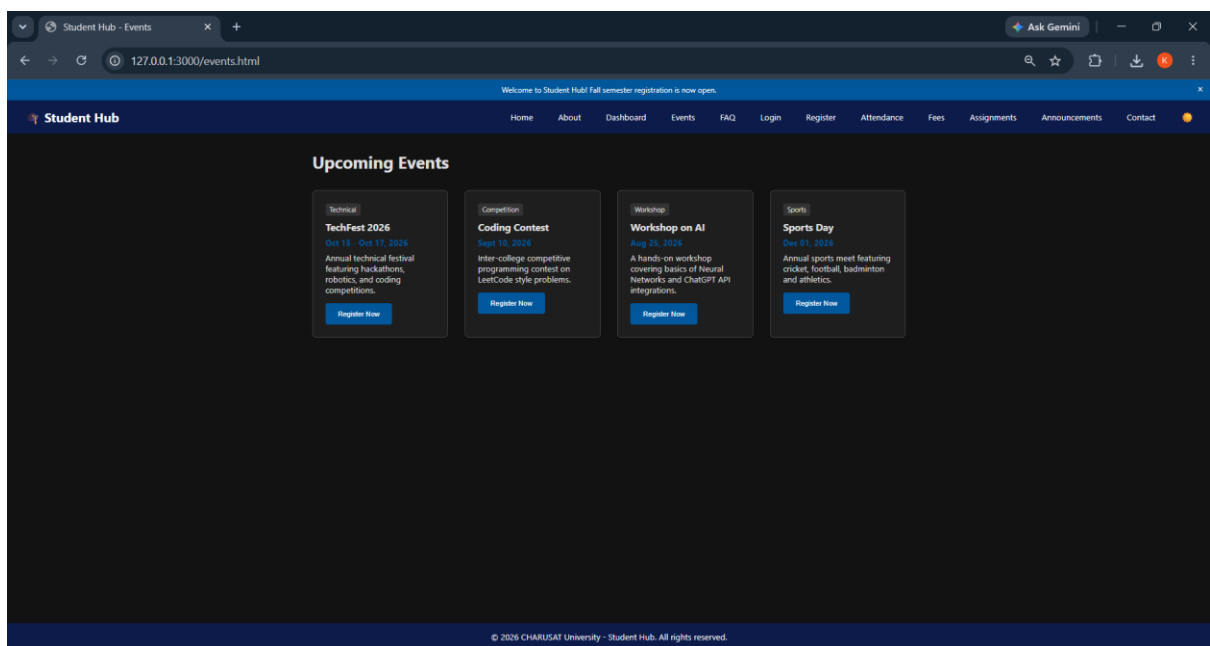
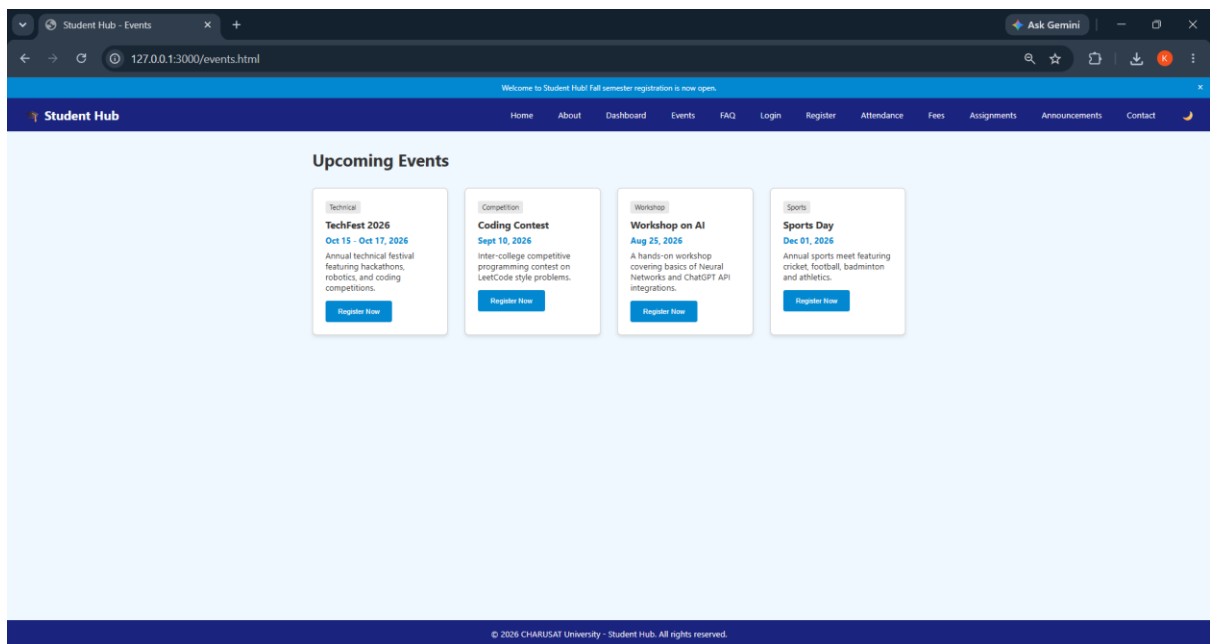
Webpages:

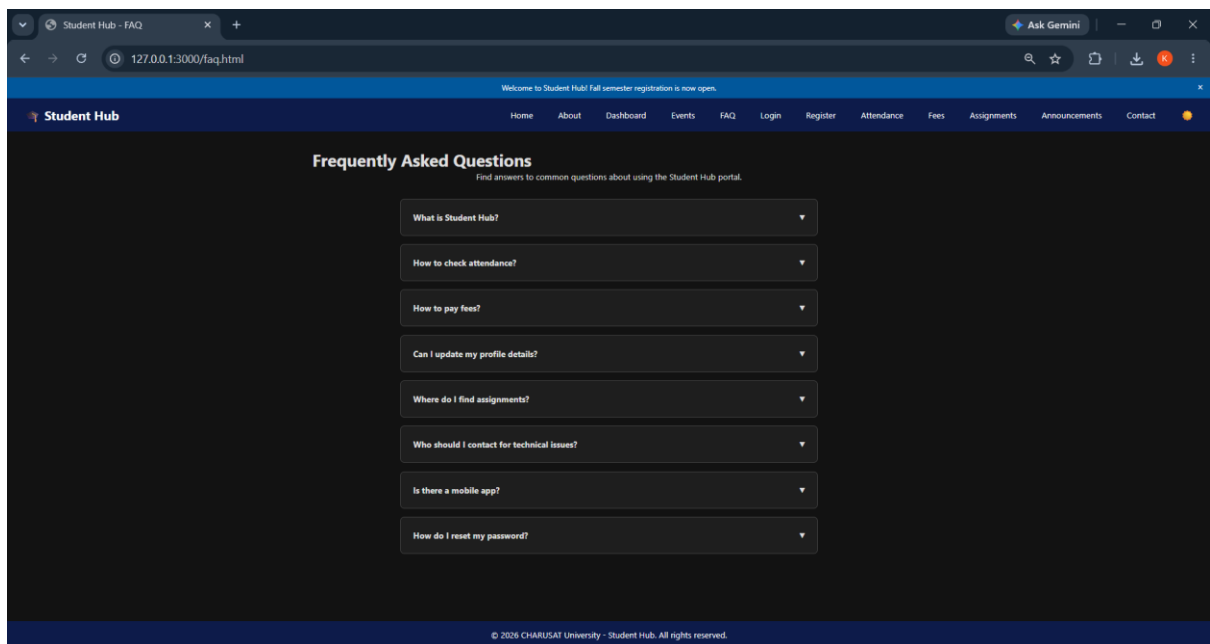
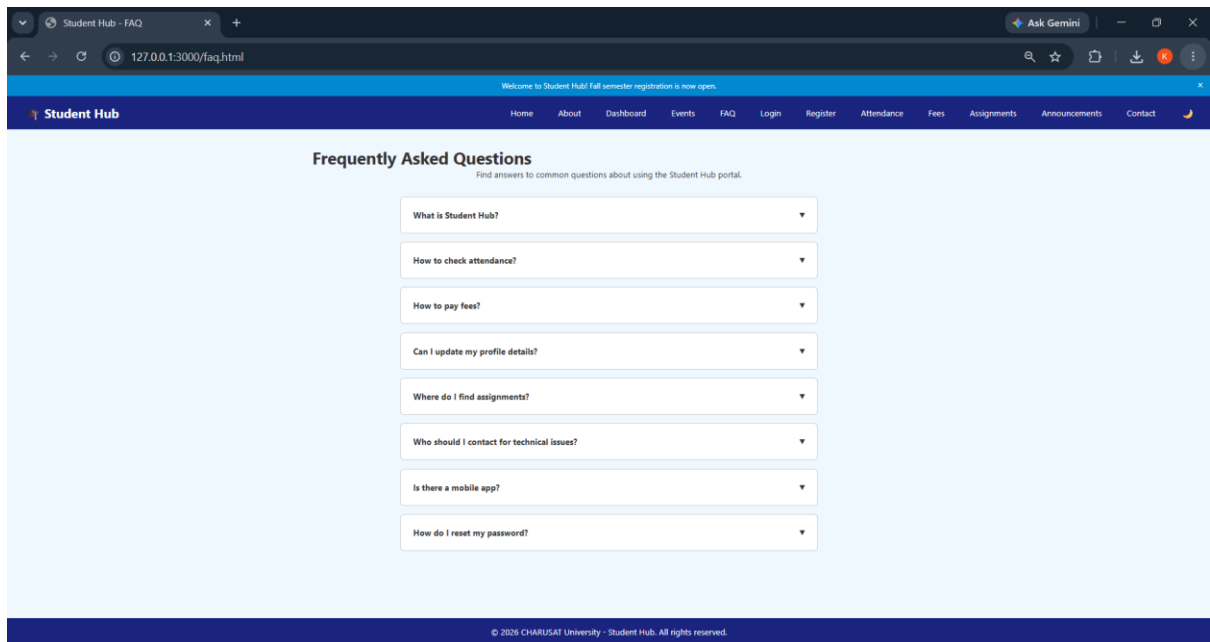


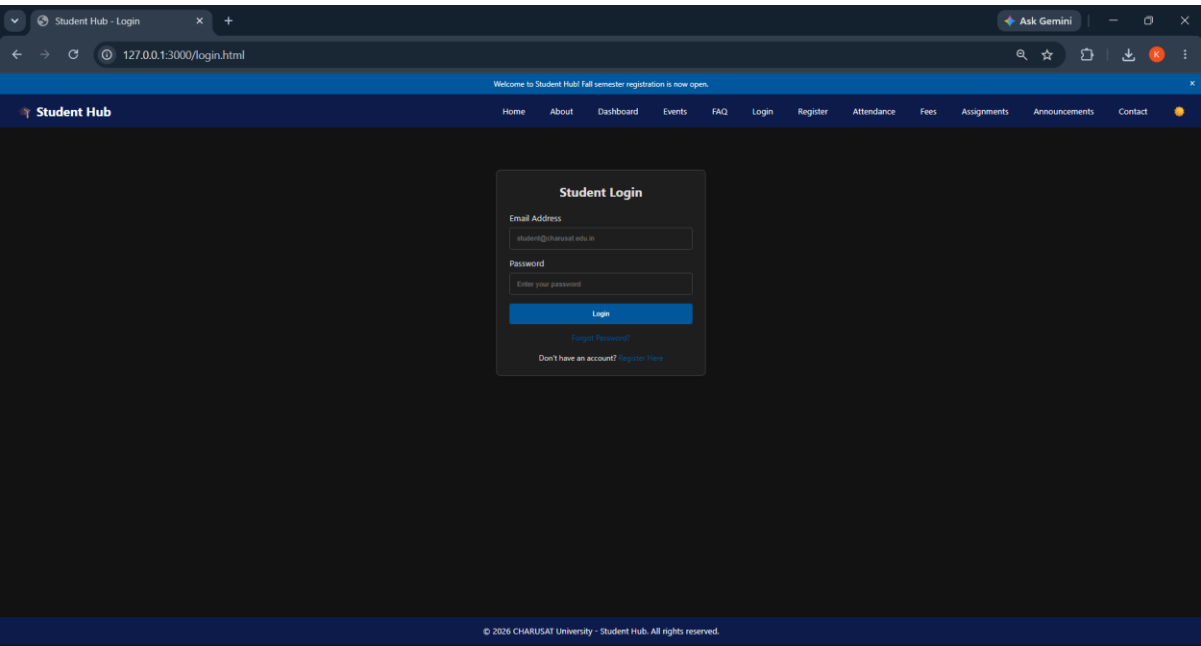
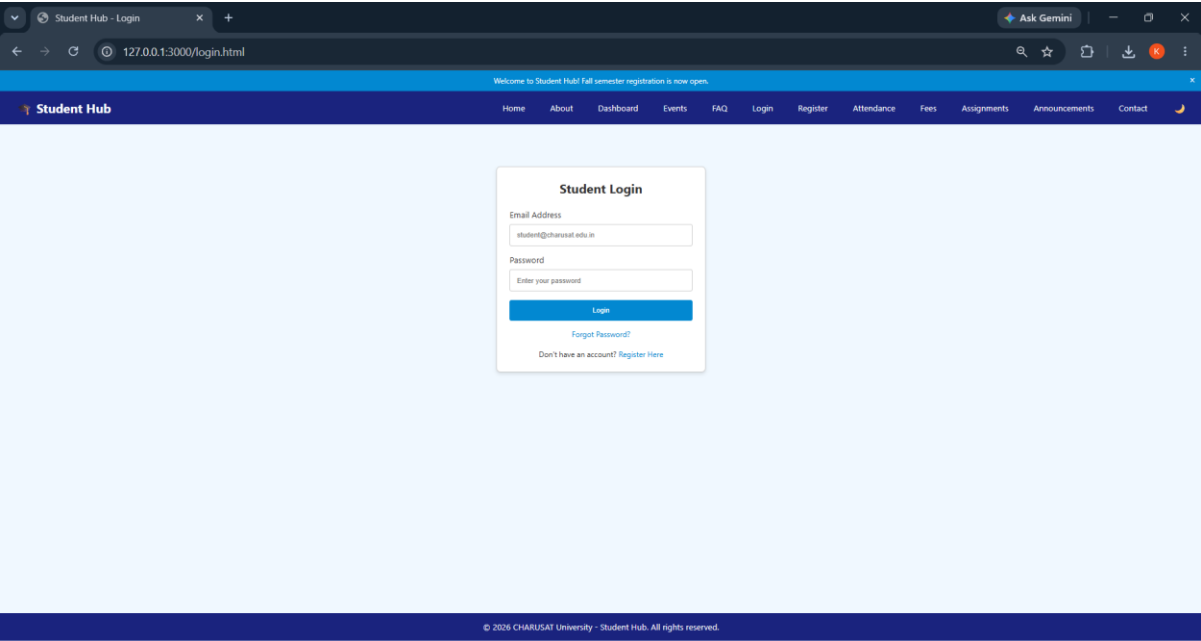


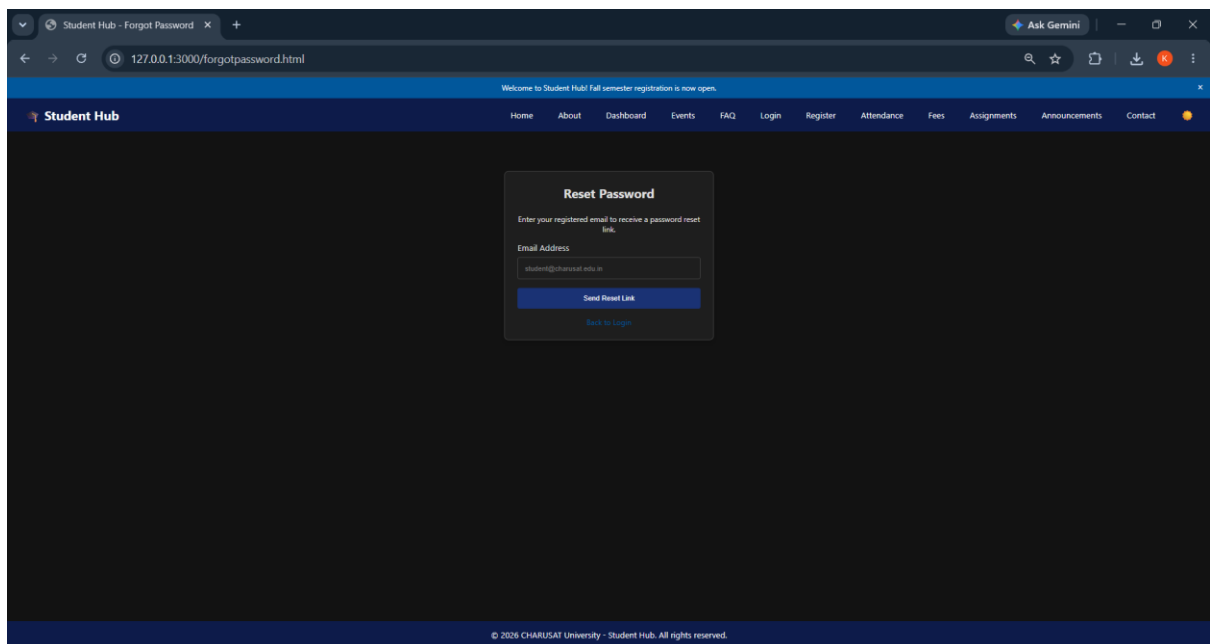
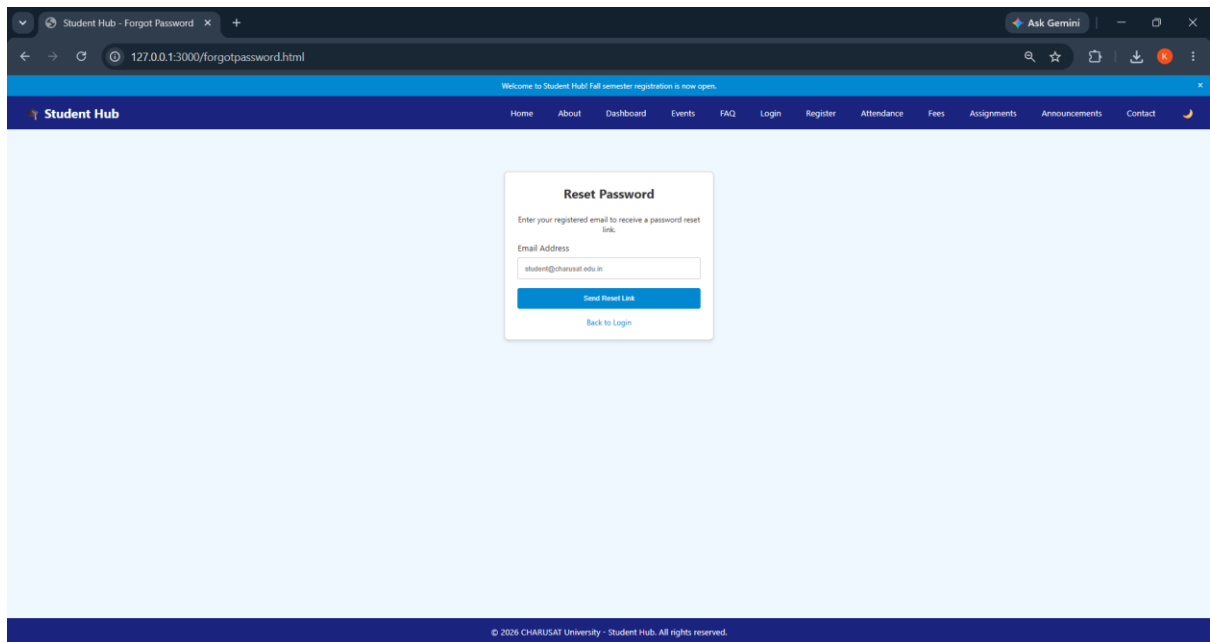


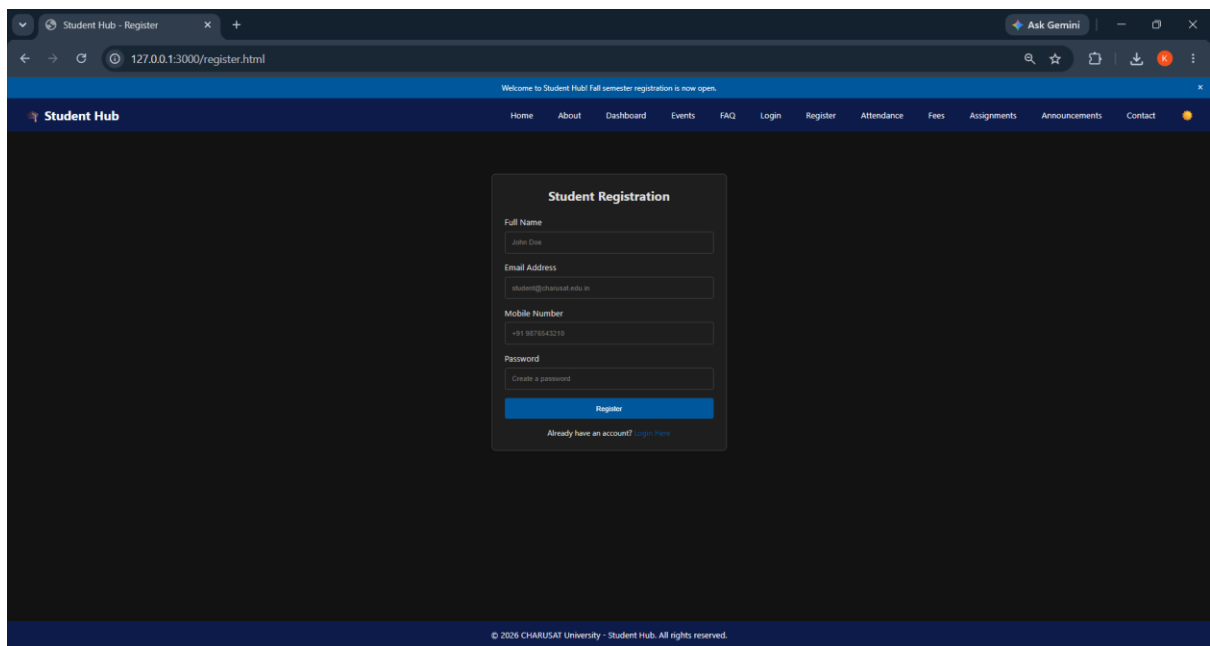
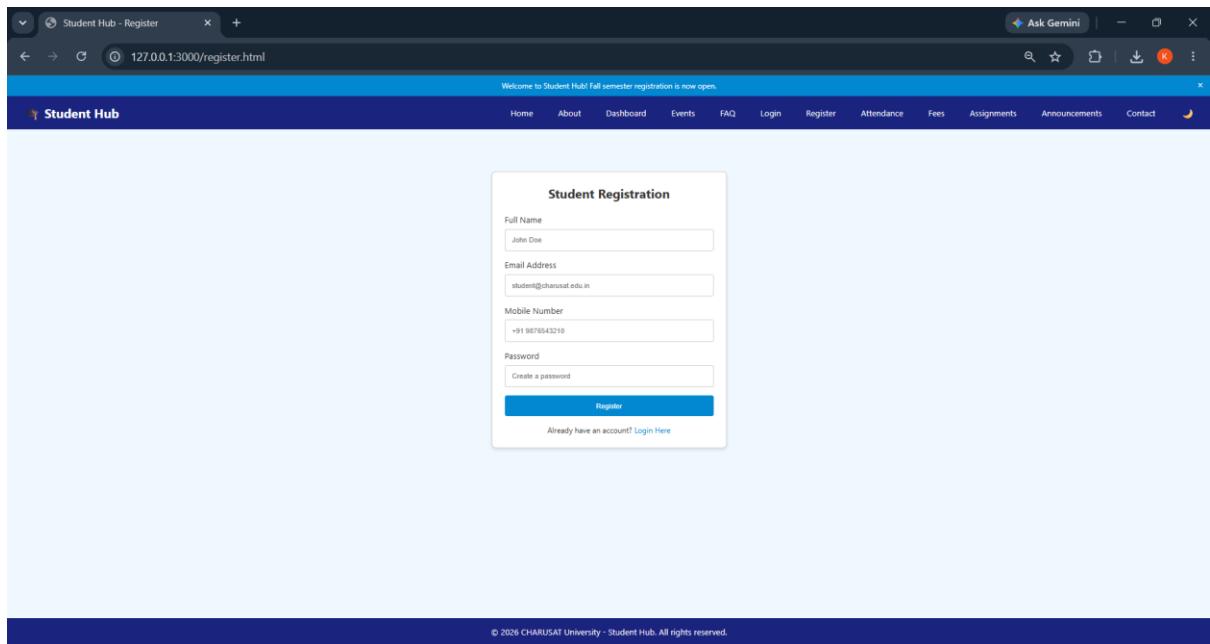












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Attendance Record

Subject Code	Subject Name	Total Classes	Attended	Percentage	Status
CS301	Web Development	40	36	90%	Good
CS302	Data Structures & Algorithms (DSA)	50	44	88%	Good
MA303	Mathematics	40	38	95%	Excellent

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Semester 3	₹65,000	July 31, 2026	Paid	Download
Semester 4	₹65,000	Dec 31, 2026	Pending	Pay Now

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